Shloke Binani

33A Canal Circular Road, Swarnamani, Kolkata-700054

+91 7003855628 | shlokebinani@gmail.com | LinkedIn | GitHub | Portfolio

EDUCATION

Vellore Institute of Technology, Vellore, India — 2023 to 2027
Bachelor of Technology in Electronics and Communication Engineering | **CGPA: 8.68 / 10.0**

Haryana Vidyamandir School, Kolkata, India — 2022 to 2023
Higher Secondary Examination (CBSE, Class XII) | **Score: 80%**

Calcutta Boys' School, Kolkata, India — 2020 to 2021
Indian Certificate of Secondary Education (ICSE, Class X) | **Score: 93.2%**

TECHNICAL SKILLS

- **Languages & Web**: Java (DSA), Python (Pandas), C++, JavaScript, React.js, SQL, C, Object-Oriented Design (OOD)
- **Core Foundations**: Data Structures & Algorithms, DBMS, Operating Systems (OS), Cloud Computing, Computer Networks (CN)
- **Backend, Cloud & Security**: FastAPI, GraphQL, PostgreSQL, MySQL, Neon, Docker, Render, Vercel, REST APIs, Identity Management (JWT), Secure Coding, Network Security & Threat Mitigation
- **AI-Assisted Development & ML**: Cursor, GitHub Copilot, Claude, Antigravity, Responsible AI & Output Validation, Machine Learning (QCNN, MLP, SVM, Random Forest, PCA, Clustering), Power BI, A/B Testing, Git, Postman, n8n
- **Soft Skills**: Agile Delivery, Cross-Functional Collaboration, Problem-Solving, Adaptability, Techno-Functional Alignment

EXPERIENCE

Kofuku Technologies Pvt. Ltd., On-Site — May 2025 - July 2025
Software Engineer Intern | Engineered scalable, secure backend solutions and intelligent workflows in an agile team

- Improved backend reliability by **30%** (cut API failures) by engineering scalable **FastAPI** and **Hasura GraphQL** services
- Resolved **JWT authentication and authorization** faults across services by enforcing **secure coding practices**
- Saved **8–12 hours/week** of manual work by automating intelligent **n8n** data pipelines to enhance team productivity
- Cut facility booking latency by **60%** by deploying an AI-powered voice agent with real-time calendar synchronization
- Reduced QA rework by **20%** in an **agile** team by standardizing API specs and automated **quality engineering** tests

ACADEMIC PROJECTS

E-Commerce Customer Analytics & Visualisation | *Power BI, SQL, Python, PCA, Clustering* — GitHub

- Executed data analytics on complex transactional logs, applying **PCA** and hybrid **DBSCAN/K-Means** to generate insights on high-value customer cohorts
- Built an executive **Power BI** Customer Lifetime Value visualisation dashboard and designed an **A/B testing** framework, delivering actionable targeting recommendations

Paradox - Scalable Ticket Booking System | *React.js, JavaScript, PostgreSQL, Render, Antigravity* — Live Link

- Accelerated full-stack delivery using **Antigravity** to build a **React.js** (**JavaScript**) frontend and **Render** backend, critically reviewing and testing all AI-generated code for security, privacy, and risk before shipping
- Prevented double-booking race conditions under high concurrency by implementing strict **row-level database locking** and **ACID-compliant** waitlist transactions on **Neon PostgreSQL**

MS-HQNet: Multi-Stream Hybrid Quantum Framework | *Python, QCNN, SM-SSA, NCA*

- Designed a Hybrid **QCNN** combining a **6-qubit** quantum circuit with a classical MLP bypass for EMG signal classification, achieving **99.17% accuracy** across **600 signals**
- Constructed robust **data engineering** pipelines utilizing **SM-SSA** and **NCA** to reduce algorithmic complexity and optimize feature extraction, ensuring system reliability

LEADERSHIP

Society of Automotive Engineers (SAE-VIT) | *Marketing, Design & PR Head* — May 2024 - May 2026

- Directed multi-channel branding campaigns reaching **1,000+ students**, collaborating with university stakeholders to expand chapter outreach
- Managed operations and **40+ volunteers** as primary POC for SAE AirShow 25, increasing attendee participation by **30 to 45%**
- Streamlined cross-functional logistics and team workflows across **12 major university events** during graVITas 25, ensuring 100% on-time execution

ACADEMIC ACHIEVEMENTS & CERTIFICATIONS

- Solved **250+ problems** across LeetCode & GeeksforGeeks (Easy to Hard), mastering Advanced Data Structures and SQL
- Awarded the **Elite Category Chapter Award** for SAE-VIT for excellence in operations, 2025 to 2026
- Awarded the **Leadership Certificate** for exceptional contribution and management by Vellore Institute of Technology
- Won **Runner-Up** at Central Hack for **ResiliNet**, mitigating 6G network congestion threats via DPI traffic routing
- Won **Runner-Up** position at Providence Leap Ideathon for AI-Driven Healthcare Benefit Orchestration

Certifications: Pandas, Kaggle; Statistics OnRamp (2024) & MATLAB OnRamp (2024), The MathWorks, Inc.